Section 1

Section A:

1. $\int x + \sqrt{x} - 2\, dx$

2. $\int (x^2 + \frac{1}{\sqrt{x}})^2\, dx$

Section B: Integration by Substitution

3. $\int \cos 5x \quad dx$

4. $\int \frac{x \quad dx}{\sin^2 5x^2}$

5. $\int \frac{1}{x} \sec^2 \ln x \;\; dx$

6. $\int \cot e^x \quad e^x \quad dx$

7. $\int e^{\sin x} \cos x\, dx$

8. $\int (e^{2x} + e^{-2x})^2\, dx$

1. $\int x + \sqrt{x} - 2\, dx$

$$\int x\, dx + \int x^{1/2}\, dx - \int 2\, dx$$

$$= \frac{x^2}{2} + \frac{x^{3/2}}{3/2} - 2x + C$$

$$= \frac{x^2}{2} + \frac{2}{3}x^{3/2} - 2x + C$$

$$\boxed{\frac{x^2}{2} + \frac{2}{3}x^{3/2} - 2x + C}$$

2. $\int \left(x^2 + \frac{1}{\sqrt{x}}\right)^2 \, dx$

First expand:

$$\left(x^2 + x^{-1/2}\right)^2 = x^4 + 2x^{2-1/2} + x^{-1}$$

$$= x^4 + 2x^{3/2} + \frac{1}{x}$$

Integrate:

$$\int x^4 \, dx = \frac{x^5}{5}$$

$$\int 2x^{3/2} \, dx = 2 \cdot \frac{x^{5/2}}{5/2} = \frac{4}{5} x^{5/2}$$

$$\int \frac{1}{x} \, dx = \ln|x|$$

Add constant C:

$$\boxed{\frac{x^5}{5} + \frac{4}{5} x^{5/2} + \ln|x| + C}$$

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3. $\int \cos 5x \, dx$

Let $u = 5x$, $du = 5\,dx \implies dx = \frac{du}{5}$.

$$\int \cos u \cdot \frac{du}{5} = \frac{1}{5} \sin u + C$$

Back substitute:

$$\boxed{\frac{1}{5} \sin 5x + C}$$

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5. $\int \frac{1}{x} \sec^2(\ln x) \, dx$

Let $u = \ln x$, $du = \frac{1}{x} dx$.

The integral becomes:

$$\int \sec^2 u \, du = \tan u + C$$

Back substitute:

$$\boxed{\tan(\ln x) + C}$$

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6. $\int \cot(e^x)\, e^x\, dx$

Let $u = e^x$, $du = e^x dx$.

$$\int \cot u\, du = \ln|\sin u| + C$$

Back substitute:

$$\boxed{\ln|\sin(e^x)| + C}$$

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7. $\int e^{\sin x} \cos x\, dx$

Let $u = \sin x$, $du = \cos x\, dx$.

$$\int e^u\, du = e^u + C$$

$$\boxed{e^{\sin x} + C}$$

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8. $\int (e^{2x} + e^{-2x})^2 \, dx$

Expand:

$$e^{4x} + 2e^{2x}e^{-2x} + e^{-4x} = e^{4x} + 2 + e^{-4x}$$

Integrate:

$$\int e^{4x} dx = \frac{e^{4x}}{4}$$

$$\int 2 \, dx = 2x$$

$$\int e^{-4x} dx = -\frac{e^{-4x}}{4}$$

Sum:

$$\boxed{\frac{e^{4x}}{4} + 2x - \frac{e^{-4x}}{4} + C}$$